Lab Experiment-9

Implementation of subnetting using cisco packet tracker.

AIM:-

To implement class less IP subnetting using cisco packet tracer

Steps:-

* open cisco packet area
* create a new network.
* Add routers, switches and PCs.
* connect devices using suitable cables.

Subnetting:-

192.168.1.0/24 using 27 gives: 8 subnet 30 usable host addresses per subnet.

configure router:-

open Router → CLI
enable
configure terminal
interface fastethernet 0/0
ip address 192.168.1.1 255.255.255.224
no shutdown
exit.

for another interface
interface fastethernet 0/1
ip address 192.168.2.1 255.255.255.224

no shutdown

exit

gigabit ethernet

enable

configure terminal

interface Gigabit ethernet 0/0

ip address 192.168.1.1 255.255.255.224

no shutdown

exit

interface fastEthernet 0/2

switchport mode access

exit

configure PC's

open:

PC → config → IP configuration

Enter:

IP address

Subnet mask: 255.255.255.224

Default gateway: Routers IP address

Test of network

1) open PC → command prompt

2) using ping cmd

3) ping another PC to router

Ex: Ping 192.168.1.1

If replies are received, network is working correctly.

Result:

Classless IP subneting was successfully